

```
!pip install mlxtend

import pandas as pd
import numpy as np
from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import apriori, association_rules

# Veriyi oku ve ilk 5 satırı gör
df = pd.read_csv('Groceries_dataset.csv')
print(f"Toplam satır sayısı: {len(df)}")
df.head()
```

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 Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.7->matplotlib>=3.
 Toplam satır sayısı: 38765

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index	Member_number ▲	Date	itemDescription
3	1187	12-12-2015	other vegetables
0	1808	21-07-2015	tropical fruit
2	2300	19-09-2015	pip fruit
1	2552	05-01-2015	whole milk
4	3037	01-02-2015	whole milk

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Next steps: [Generate code with df](#) [New interactive sheet](#)

```
# 2. Aynı müşteri ve aynı tarihe ait ürünleri tek bir sepet(transaction) listesinde toplayalım
transactions = df.groupby(['Member_number', 'Date'])['itemDescription'].apply(lambda x: list(set(x))).tolist()

print(f"Toplam Sepet (transaction) sayısı: {len(transactions)}")
print(f"Örnek İlk 3 Sepet: \n{transactions[:3]}")
```

Toplam Sepet (transaction) sayısı: 14963

Örnek İlk 3 Sepet:

[['semi-finished bread', 'whole milk', 'yogurt', 'sausage'], ['pastry', 'salty snack', 'whole milk'], ['canned beer', 'misc.

```
# 3. TransactionEncoder ile sepetleri ikili matrise (True/False) dönüştürelim
te = TransactionEncoder()
te_ary = te.fit(transactions).transform(transactions)
df_transformed = pd.DataFrame(te_ary, columns=te.columns_)
```

```
print(f"Matris Boyutu (Sepet x Ürün Sayısı): {df_transformed.shape}")
df_transformed.head()
```

Matris Boyutu (Sepet x Ürün Sayısı): (14963, 167)

	Instant food products	UHT- milk	abrasive cleaner	artif. sweetener	baby cosmetics	bags	baking powder	bathroom cleaner	beef	berries	beverages	bottled beer	bottled water	brand
0	False	False	False	False	False	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False	False	False	False	False

5 rows x 167 columns

```
# 4. En az %0.52 sıklıkla birlikte görülen ürün gruplarını buluyoruz
frequent_itemsets = apriori(df_transformed, min_support=0.0052, use_colnames=True)

# Kaç adet sık ürün kümesi bulunduğunu yazdıralım
print(f"Bulunan Sık Ürün Kümesi Sayısı: {len(frequent_itemsets)}")
frequent_itemsets.sort_values(by='support', ascending=False).head(10)
```

Bulunan Sık Ürün Kümesi Sayısı: 123

1 to 10 of 10 entries Filter ?

index	support	itemsets
86	0.15792287642852368	frozenset({'whole milk'})
52	0.12210118291786407	frozenset({'other vegetables'})
65	0.11000467820624206	frozenset({'rolls/buns'})
74	0.09710619528169484	frozenset({'soda'})
87	0.08587850030074183	frozenset({'yogurt'})
66	0.06957160997126245	frozenset({'root vegetables'})
80	0.0677671589921807	frozenset({'tropical fruit'})
6	0.06068301811134131	frozenset({'bottled water'})
69	0.06034886052262247	frozenset({'sausage'})
19	0.05313105660629553	frozenset({'citrus fruit'})

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```
# 5. Güven (confidence) eşiği 0.12 olan kuralları çıkaralım
rules = association_rules(frequent_itemsets, metric="confidence", min_threshold=0.12)

# Kuralları confidence değerine göre büyükten küçüğe sıralayıp ilk 5 tanesini seçelim
top_5_rules = rules.sort_values(by='confidence', ascending=False).head(5)

# Çıktıyı temiz ve okunabilir bir tablo olarak formatlayalım
result_table = top_5_rules[['antecedents', 'consequents', 'support', 'confidence', 'lift']].copy()
result_table['antecedents'] = result_table['antecedents'].apply(lambda x: ', '.join(list(x)))
result_table['consequents'] = result_table['consequents'].apply(lambda x: ', '.join(list(x)))
result_table.columns = ['Öncül Ürün (X)', 'Ardıl Ürün (Y)', 'Support (Destek)', 'Confidence (Güven)', 'Lift(Kaldıraç)']

print("--- CONFIDENCE DEĞERİ EN YÜKSEK İLK 5 BİRLİKTELİK KURALI ---")
result_table.reset_index(drop=True)
```

--- CONFIDENCE DEĞERİ EN YÜKSEK İLK 5 BİRLİKTELİK KURALI ---

	Öncül Ürün (X)	Ardıl Ürün (Y)	Support (Destek)	Confidence (Güven)	Lift(Kaldıraç)
0	bottled beer	whole milk	0.007151	0.157817	0.999330
1	sausage	whole milk	0.008955	0.148394	0.939663
2	newspapers	whole milk	0.005614	0.144330	0.913926
3	domestic eggs	whole milk	0.005280	0.142342	0.901341
4	frankfurter	whole milk	0.005280	0.139823	0.885388